Capstone Project2

HEART DISEASE DETECTION

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INTRODUCTION

Heart disease is one of the leading causes of death worldwide, making early detection extremely important. Accurate prediction helps doctors take timely decisions, reduce risks, and improve patient outcomes.

In this project, we aim to predict the presence of heart disease using patient clinical and diagnostic data.

The dataset includes important attributes such as age, blood pressure, cholesterol levels, ECG results, maximum heart rate, and exercise-induced angina. Using these features, we apply machine learning techniques to build a model that can assist in early diagnosis of heart disease.

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Problem Statement

Primary Question:

Can we accurately predict the presence of heart disease using a patient's clinical and diagnostic test data?

Business Context:

Heart disease is one of the leading causes of death worldwide. Early detection through data-driven prediction can improve patient outcomes and reduce healthcare costs. This project represents a real-world scenario where predictive models assist doctors in clinical decision-making.

Challenge:

Based on medical attributes such as blood pressure, cholesterol, ECG results, and exercise-related measurements, build a model to classify whether a patient has heart disease or not.

Overview of the Project

Objective:

Develop machine learning models to predict heart disease
Use patient clinical and diagnostic data
Improve early detection and risk assessment

Scope:

Analyze medical attributes such as blood pressure, cholesterol, ECG results, and exercise capacity
Build classification models
Support healthcare decision-making

Approach:

Data exploration and preprocessing
Model development (Decision Tree, Random Forest, Logistic Regression, SVM)
Hyperparameter tuning using grid search and cross-validation
Model Evaluation: Evaluate using accuracy, precision, recall, F1-score, ROC-AUC
Feature importance analysis

Outcome:

Accurate heart disease prediction model
Identification of key risk factors

Data generation & Loading

The project uses a comprehensive heart disease dataset sourced from publicly available clinical repositories.

It includes 400 patient records, each with 13 clinical features such as age, sex, cholesterol, blood pressure, and ECG results.

Data acquisition ensured no missing values, enabling a smoother preparation process.

The dataset was loaded into the environment using Python's Pandas library, allowing for efficient manipulation and analysis.

Initial data inspection confirmed correct data types and consistent formatting for all features.

Efficient data loading and validation lay the groundwork for subsequent data exploration and modeling

Data Exploration & Understanding

- The dataset includes 13 features (age, sex, chest pain type, blood pressure, cholesterol, ECG, etc.), providing a broad view of heart health indicators.

- No missing values, stable ranges—e.g., age from 29 to 77, cholesterol from 120 to 564 mg/dl.

- Visual exploration revealed most patients are male and chest pain type 'asymptomatic' is most common.

- Key categorical and numerical features distributed evenly—ready for advanced analysis.

Exploratory Data Analysis

1. Purpose of EDA:

- Understand the dataset and how features relate to heart disease
- Spot unusual values or outliers

2. Target Variable Analysis:

- Check how many patients have heart disease (1) and how many don't (0)

3. Feature Analysis:

- Numerical features: age, cholesterol, blood pressure, max heart rate, ST depression
- Categorical features: chest pain type, thalassemia, resting ECG, exercise-induced angina

4. Correlation Analysis:

- Checked which features are strongly related to heart disease.

5. Visuals to Show:

- Bar chart of target variable
- Heatmap of feature correlations
- Boxplots/histograms of numerical features

Data Preprocessing & Feature Engineering

- Handle Missing Values: No missing data found
- Encode Categorical Variables: Converted chest pain, ECG, thal, etc. into numbers
- Feature Scaling: Standardized cholesterol, BP, max heart rate
- Data Splitting: 80% training, 20% testing
- Feature Selection: Chose important features based on correlation

Model Building & Training

Step 1: Initialize Models
Decision Tree
Random Forest
Logistic Regression
Support Vector Machine

Step 2: Train Models
Split dataset: 80% training, 20% testing
X_train → patient features (age, blood pressure, cholesterol…)
y_train → labels (0 = No disease, 1 = Heart disease)

Step 3: Make Predictions
y_pred = model.predict(X_test) → Predict yes/no for new patients
y_prob = model.predict_proba(X_test)[:,1] → Gives probability of heart disease

Step 4: Hyperparameter Tuning
Use GridSearchCV to find the best set of model parameters
Models retrained with optimal parameters → improved accuracy and reliability

Model Evaluation & Performance Analysis

Step 1: Evaluation Metrics
Accuracy → How many predictions were correct
Precision → How many predicted heart disease cases were actually correct
Recall → How many actual heart disease cases were identified
F1-Score → Balance between Precision & Recall
ROC-AUC → Measures model's ability to rank patients from least to most likely to have disease

Step 2: Confusion Matrix
Shows True Positives, True Negatives, False Positives, False Negatives
Helps understand where the model is making mistakes

Step 3: Feature Importance
Decision Tree & Random Forest → Identify which features most impact predictions
Example: Chest pain type, maximum heart rate, cholesterol
Step 4: Performance Summary
Compare all models on metrics above
Best performing model chosen based on ROC-AUC and F1-Score

Stream API Deployment

Build Python apps quickly with app.py.

Run locally: streamlit run app.py.

Deploy on: Streamlit Cloud, Heroku, or AWS/GCP/Azure.

Pros: Easy, fast setup, no frontend needed.

Cons: Limited customization, not for heavy APIs.

Docker Deployment

Package app + dependencies in containers.

Steps: Write Dockerfile → docker build → docker run.

Deploy via Docker Hub, AWS ECS, GCP, Azure, or Kubernetes.

Pros: Portable, scalable, consistent environment.

Cons: Learning curve, needs orchestration at scale.

Conclusion

In conclusion, our capstone project demonstrated the powerful role of machine learning in healthcare. By applying classification algorithms to clinical diagnostic data, we were able to achieve strong predictive performance in identifying patients at risk of heart disease.

Key insights include:

- Early and accurate prediction of heart disease is feasible with machine learning.
- Certain features, such as cholesterol, chest pain type, and maximum heart rate, play a critical role in prediction.
- Optimized models can serve as a decision-support tool for healthcare providers, improving efficiency and patient outcomes.